Akhrorjon Rakhmonov

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Summary

I am an experienced AI Researcher and Engineer with over 3 years of dedicated practice in the fields of computer vision (CV) and natural language processing (NLP). My journey in artificial intelligence has been driven by a passion for innovation and a commitment to leveraging advanced technologies to solve complex problems. I have a proven track record of developing and deploying state-of-the-art AI solutions that enhance operational efficiency and deliver impactful results across various industries. My research work and real-world projects can be explored in more detail in my Google Scholar profile and online portfolio.

Skills

- Highly proficient in Python
- Extremely experienced at Deep Learning frameworks, such as PyTorch, PyTorch Lightning, transformers, and Hugging Face
- Skilled at Deep Learning frameworks, such as TensorFlow, Keras, and NumPy
- Exceptionally adept at Machine Learning libraries, such as scikit-learn, cv2, pandas, and scipy
- Capable of server and cloud technologies, such as docker, API development as well as version control systems such as Git
- Skilled in communication & teamwork and adept at working under pressure

Education

Kyungpook National University — Daegu, South Korea
PhD in Computer Science and Engineering — *GPA - 4.42/4.5*

Keimyung University — Daegu, South Korea
Global Masters in Business Administration — *GPA - 4.34/4.5*

Experience

- **Principal AI Researcher** — OnD AI, South Korea
  Road Accident Fault Judgment AI System Using Blackbox Videos — *November, 2025 - Ongoing*
  - **DB Insurance Project Leader**:
    - Leading a research on development of AI System to predict fault judgment in road accidents using videos from blackbox cameras with DB Insurance Company
    - Responsible for development of AI method for accurate prediction of fault judgment of road accidents
    - Responsible for development of long-term and short-term plans of the project as well as task allocation for team members
    - Continuously optimize the accuracy and speed of the AI system to meet customer needs

- **Lecturer** — Kyungpook National University, South Korea
  Software Education Institute — *March, 2025 - August, 2025*
  - **Python Programming**:
    - Taught basics of Python Programming Language to undergraduate students of Kyungpook National University
    - Prepared interactive coding problems and conducted practical exercises
    - Prepared practical assignments covering the classes' topics
    - Graded the students based on exams and assignment results

- **Graduate School Researcher** Kyungpook National University, South Korea
  Multimedia Processing Information Lab AI Lead *September, 2022 - August, 2025*
  - **Computer Vision**:
    * Led and successfully executed 5+ long-term projects in computer vision
    * Specialized in image classification, semantic segmentation, object detection tasks
    * Published 5+ scientific papers in the most prestigious SCIE journals based on the conducted projects
    * Instructed undergraduate students coding practice by facilitating their proficiency in deep learning frameworks, such as TensorFlow and PyTorch
    * Supervised and guided undergraduate students in writing scientific papers on deep learning methodologies and applications, fostering their research and presentation skills
    * Cooperated with industrial companies to integrate AI technologies into their production processes, leveraging expertise in computer vision to enhance efficiency and innovation.
  - **Natural Language Processing**:
    * Accomplished several sequence data-based projects, including text classification and sentiment analysis
    * Collaborated closely with project managers to ensure adherence to project timelines and milestones, facilitating the timely delivery of AI projects
    * Instructed undergraduate students coding practice by facilitating their proficiency in NLP-related deep learning frameworks, such as PyTorch, transformers, and Hugging Face

Projects & Research

- **A Hybrid Unsupervised-Weakly Supervised Method for Video Anomaly Detection**:
  - Hybrid-ConvATFM is a novel video anomaly detection method that operates in both unsupervised and weakly supervised settings by using GMM-based clustering to generate pseudo-labels when annotations are unavailable. The method combines convolutional autoencoders with attention modules to capture complex temporal dynamics and incorporates temporal feature magnitude learning for precise anomaly classification. In weakly supervised and fully supervised settings, the proposed method delivers performance that outperforms its SOTA counterparts.

  Experience: Python, PyTorch, Weakly Supervised, Video Anomaly Detection
- **Knowledge Distillation-based Efficient Industrial Anomaly Detection**:
  - In this paper, we undertake research focused on the utilization of knowledge distillation technique to classify defects in industrial products in an efficient way by distilling knowledge from an experienced teacher model to a lighter student model. Owing to the contributions, the proposed method outperforms its SOTA counterparts in several benchmarks.

  Experience: Python, PyTorch, Knowledge Distillation, Image Anomaly Detection
- **Attention-based Network with Optional Activation Function for Unsupervised Video Anomaly Detection**:
  - In this paper, we proposed a robust, efficient, and fast network (AONet) that employs a future frame prediction method to detect unusual events in video sequences. Inputs of several previous frames are fed into an autoencoder-based model with skip connections and attention mechanism to learn deep spatial and temporal features of input frames to predict a next frame. In addition, we introduced a robust activation function called OptAF that combines the advantages of popular activation functions such as ReLU, Leaky ReLU and Sigmoid.

  Experience: Pytorch, Efficient Computation, Model Architecture, Video Anomaly Detection
- **AED-Net: Attention-Based Detection Model for Disabled Signage Detection**:
  - In this study we propose to make certain modifications to a baseline YOLOv5 model. Instead of the original 9 blocks in the backbone and 4 C3 blocks, we propose to replace them with 6 and 4 EfficientNet blocks, accordingly. We have also incorporated an attention mechanism into the proposed architecture before the detection phase. With these enhancements, a novel object detector, attention-based efficient detection model (AED-Net) is proposed. AED-Net shows superior performance compared to the baseline method on a custom dataset comprising images of cars displaying disabled signage on their windshields.

  Experience: PyTorch, Efficientnet, Object Detection, YOLOv5, Attention Mechanism

- **Enforced Clustering for Zero-to-One-Shot Texture Anomaly Detection**:
  - This paper focuses on homogeneous textures and demonstrates how the problem can be addressed without any training samples or additional training (zero-shot), only requiring one normal sample (one-shot) for hyperparameter selection, which is an additional challenge in unsupervised settings. Experiments show that this zero-to-one-shot method, which facilitates deployment by reducing data dependency, maintains performance comparable to, or even higher than, conventional many-shot methods, all with relatively high speed.

  Experience: Python, PyTorch, Anomaly Detection, Zero-to-One-Shot Learning
- **Consecutive Multiscale Feature Learning-based Image Classification Model**:
  - This paper presents a consecutive multiscale feature-learning network (CMSFL-Net) that employs a consecutive feature-learning approach based on the usage of various feature maps with different receptive fields to achieve faster training/inference and higher accuracy. In the conducted experiments using six real-life image classification datasets, including small-scale, large-scale, and limited data, the CMSFL-Net exhibits an accuracy comparable with those of existing SOTA efficient networks. Moreover, the proposed system outperforms them in terms of efficiency and speed and achieves the best results in accuracy-efficiency trade-off.

  Experience: Python, PyTorch, Image Classification, Receptive Field
- **Medical Image Segmentation**:
  - In this project, we implement image segmentation using original UNet model built and trained from scratch on a custom dataset of cardiac disease images. This project also features a pretrained model called Segformer from semantic-models-pytorch library using fine-tuning. All results are shown using mIoU and PA metrics along with generated masks out of the model. Additionally, we provide an onnx converter file and deployment demo file in a streamlit cloud.

  Experience: Python, PyTorch, Medical Image Segmentation, UNet, Streamlit, ONNX
- **Various ML and LLM Projects**:
  - Successfully implemented different ML and LLM projects with real-life datasets, featuring Regression, Clustering, Dimensionality Reduction, LLMs with RAGs and Agents.

  Experience: Python, PyTorch, Jupyter Notebook, Scikit Learn, LLM

Info & Languages

Gender	Male
Age, DOB	34 years, 1992/03/16
Nationality	Uzbekistan
English	TOEIC (945);
Korean	TOPIK (Level 6);
Uzbek	Native
Russian	Bilingual proficiency